CPEN 416

Lecture 17: manipulating density matrices; quantum channels

Thursday 12 November 2026

Announcements

- Remember to fill out weekly project surveys
- Tutorial Monday: **change of plans**, practice for observables / expectation values; next tutorial assignment week after
- Quiz 8 Tuesday

Last time

We introduced observables

$$M = M^\dagger$$

(Hermitian)

Ex: $M = Z$

eigvecs: $|0\rangle, |1\rangle$

eigvals: $+1, -1$

We measure an observable by associating its eigenvectors with a basis and assigning real values (its eigenvalues) as outcomes.

The *expectation value* of an observable is

$$\langle M \rangle = \langle \psi | M | \psi \rangle$$

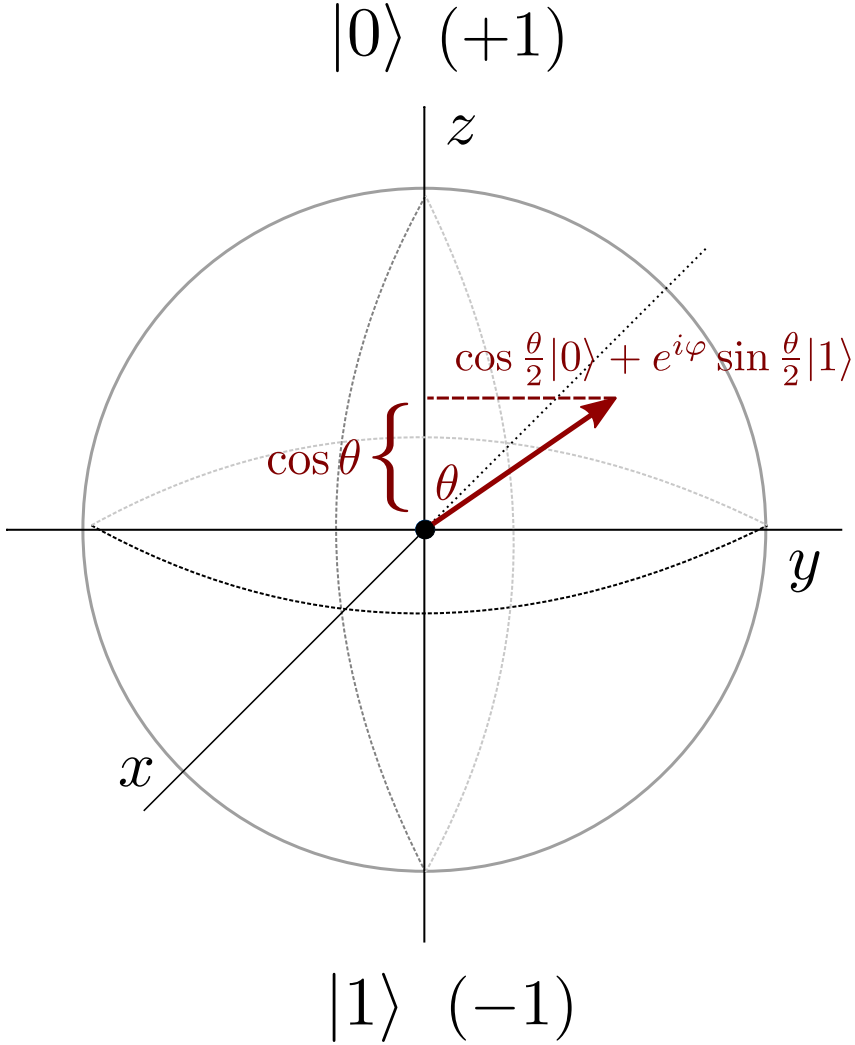
by spectral thm, $M = \sum \lambda_i |v_i\rangle\langle v_i|$

$= \sum \lambda_i \cdot \text{Prob}(\text{eig vec. } i)$

(weighted average)

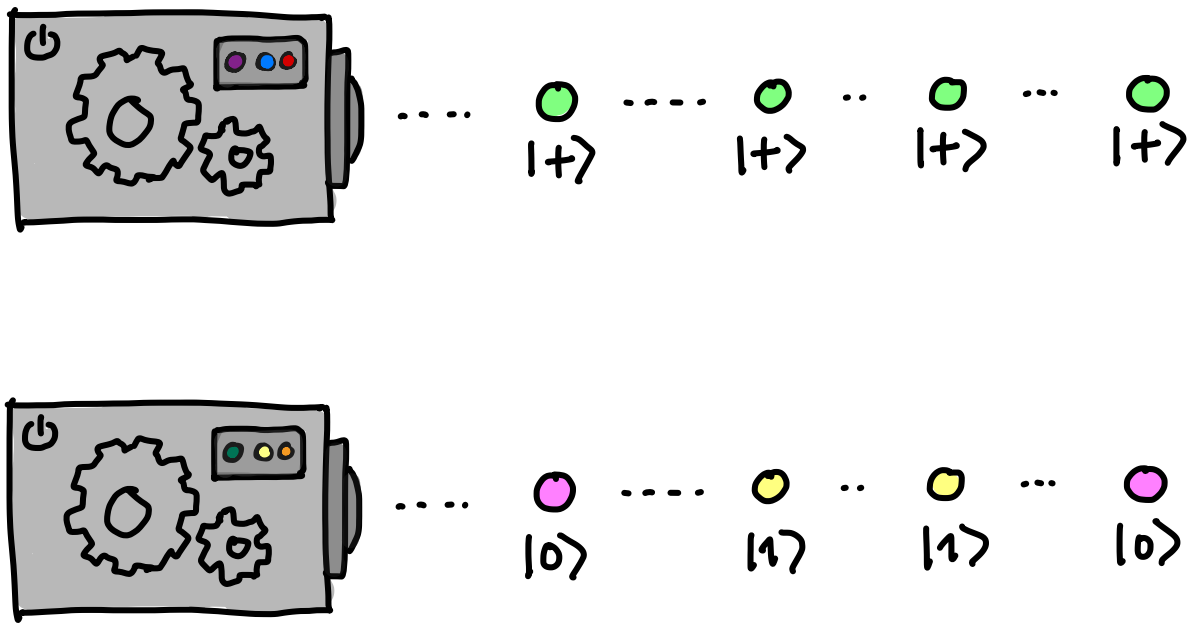
Last time

We associated expectation values of Paulis to the Bloch vector.



Last time

We introduced *mixed states*.



Density matrices

For one pure state $|\psi\rangle$, $\rho = |\psi\rangle\langle\psi|$

Mixed states are represented by density matrices,

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i| \quad , \quad \sum_i p_i = 1$$

Important properties of density matrices:

- they are Hermitian
- they have trace 1
- they are positive semi-definite (all eigenvalues are ≥ 0)
- (for pure states only) they are projectors, i.e., $\rho^2 = \rho$

Learning outcomes

- Apply unitary operations to mixed states
- Perform measurements on mixed states
- Define and apply quantum channels to quantum states

Unitary operations and mixed states

$$|+\rangle\langle+| = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}\cdot\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\end{pmatrix} = \frac{1}{2}\begin{pmatrix}1 & 1\\1 & 1\end{pmatrix}$$

Review exercise: A system prepares $|+\rangle$ with probability $1/3$, and $|0\rangle$ with probability $2/3$. What is its state?

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$$

$$= \frac{1}{3}|+\rangle\langle+| + \frac{2}{3}|0\rangle\langle 0|$$

$$= \frac{1}{6}\begin{pmatrix}1 & 1\\1 & 1\end{pmatrix} + \frac{4}{6}\begin{pmatrix}1 & 0\\0 & 0\end{pmatrix}$$

$$= \begin{pmatrix}5/6 & 1/6\\1/6 & 1/6\end{pmatrix}$$

$|-\rangle\langle-| = \frac{1}{2}\begin{pmatrix}1 & -1\\-1 & 1\end{pmatrix}$

Unitary operations and mixed states

is applied

Exercise: if H is applied to our mixed state ($|+\rangle$ w/prob. 1/3, $|0\rangle$ w/prob. 2/3), what is the new state?

original $\rho = \frac{1}{3}|+\rangle\langle+| + \frac{2}{3}|0\rangle\langle 0|$

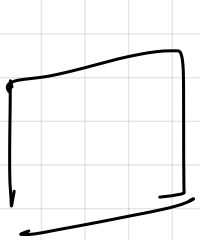
after H: $\rho' = \frac{1}{3}|0\rangle\langle 0| + \frac{2}{3}|+\rangle\langle+|$

$$= \frac{1}{3}\begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix} + \frac{2}{3}\cdot\frac{1}{2}\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}$$

$$= \begin{pmatrix}2/3 & 1/3\\ 1/3 & 1/3\end{pmatrix}$$

Mixed state example:

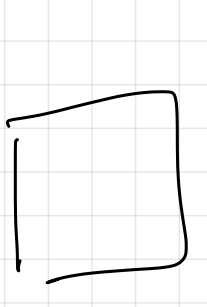
pure:



$|\Psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

(superposition, and entangled but still pure)

mixed:



1/4 $|\Psi_{00}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

1/4 $|\Psi_{01}\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$

⋮

mixture

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after H: $\rho' = \frac{1}{3}|0\rangle\langle 0| + \frac{2}{3}|+\rangle\langle +|$

$= \frac{1}{3}\left(H|+\rangle\right)\left(\langle +|H^\dagger\right) + \frac{2}{3}\left(H|0\rangle\right)\left(\langle 0|H^\dagger\right)$

$= H\left[\frac{1}{3}|+\rangle\langle +| + \frac{2}{3}|0\rangle\langle 0|\right]H^\dagger$

$= H\rho H^\dagger$

↖ original state

Unitary operations and mixed states

For pure state $|\psi\rangle$ and operation U,

$$|\psi\rangle \longrightarrow |\psi'\rangle = U|\psi\rangle$$

As mixed states,

$$|\psi\rangle\langle\psi| \longrightarrow |\psi'\rangle\langle\psi'| = U|\psi\rangle\langle\psi|U^\dagger$$

$$\rho' = U\rho U^\dagger$$

More generally,

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$$

$$\rho' = U\left(\sum_i p_i |\psi_i\rangle\langle\psi_i|\right)U^\dagger = U\rho U^\dagger$$

$$= \sum_i p_i U|\psi_i\rangle\langle\psi_i|U^\dagger = \sum_i p_i |\psi_i'\rangle\langle\psi_i'|$$

Mixed states and measurements

Recall that for a pure state $|\psi\rangle$, the probability of measuring and observing it in state $|\varphi\rangle$ is

$$\Pr(\varphi) = |\langle\varphi|\psi\rangle|^2$$

We can rewrite this...

$$= \langle\varphi|\psi\rangle\langle\psi|\varphi\rangle$$
$$= \langle\psi|\varphi\rangle\langle\varphi|\psi\rangle$$
$$= \langle\psi|\left(|\varphi\rangle\langle\varphi|\right)|\psi\rangle$$
$$= \langle\psi|\,\Pi_\varphi\,|\psi\rangle$$

$\langle Z\rangle = \langle\psi|Z|\psi\rangle$

$= \langle\psi|\cdot\lambda_0|0\rangle\langle 0|\,|\psi\rangle$

$+\langle\psi|\lambda_1|1\rangle\langle 1|\,|\psi\rangle$

$= \lambda_0\langle\psi|\Pi_0|\psi\rangle$

$+\lambda_1\langle\psi|\Pi_1|\psi\rangle$

$= \lambda_0 \cdot \text{prob}(0)$

$+\lambda_1 \cdot \text{prob}(1)$

$\Pi_\varphi = |\varphi\rangle\langle\varphi|$ is the density matrix of $|\varphi\rangle$, which is a *projector*.

Mixed states and measurements

projective meas: basis states $\{|k\rangle\}$

For mixed states, measurement follows the **Born rule**:

$$\Pr(\text{outcome } i) = \mathrm{Tr}\left(\Pi_i \rho\right)$$

where $\{\Pi_i\}$ is a **positive operator-valued measure (POVM)**.

The POVM elements satisfy

$$\sum_i \Pi_i = \mathbb{1}$$

Exercise: show this is equivalent to a projective measurement for pure states. $\mathrm{Tr}[\Pi_i \rho) \to |\langle \varphi_i | \psi \rangle|^2$ (try it!)

Mixed states and measurements

Example: $\{|+\rangle\langle+|, |-\rangle\langle-|\} = P_H$

$\Pi_+ \quad \Pi_-$

Given a state ρ, $\rho = |\psi\rangle\langle\psi| \Rightarrow \text{Prob}(+) = \text{Tr}(\Pi_+ \rho)$

$= \text{Tr}(|+\rangle\langle+| \cdot |\psi\rangle\langle\psi|)$

$= \cdots$

$= |\langle+|\psi\rangle|^2$

For general ρ (not pure),

$\Pr(+) = \text{Tr}(\Pi_+ \rho)$

$\Pr(-) = \text{Tr}(\Pi_- \rho)$

Exercise: Show that $\{|+\rangle\langle+|, |-\rangle\langle-|\}$ form a legit POVM.

$$\sum_i \Pi_i = \mathbb{1}$$

Mixed states and measurements

Exercise: Suppose we prepare our system in $|+\rangle$ with probability 1/3 and $|0\rangle$ with probability 2/3. What is the probability of obtaining the POVM outcome $\Pi_+ = |+\rangle\langle+|$? when meas. P_H.

Intuition:

1/3 time $|+\rangle$: get $|+\rangle$

2/3 time $|0\rangle$: get $|0\rangle$ with prob 1/2

$|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$

final $\frac{1}{3} + \frac{2}{3}\cdot\frac{1}{2} = \frac{2}{3}$

Math:

$$\text{Prob}(+) = \text{Tr}(|+\rangle\langle+| \cdot \rho)$$
$$= \text{Tr}\left(|+\rangle\langle+| \cdot \left(\tfrac{1}{3}|+\rangle\langle+| + \tfrac{2}{3}|0\rangle\langle 0|\right)\right)$$
$$= \text{Tr}\left(\tfrac{1}{3}|+\rangle\langle+|+ \tfrac{2}{3}\langle+|0\rangle \cdot |+\rangle\langle 0|\right)$$
$$= \text{Tr}\left(\tfrac{1}{3}|+\rangle\langle+| + \tfrac{2}{3}\cdot\tfrac{1}{\sqrt{2}}|+\rangle\langle 0|\right)$$
$$= \frac{1}{3}\text{Tr}(|+\rangle\langle+|) + \frac{2}{3}\cdot\frac{1}{\sqrt{2}}\text{Tr}(|+\rangle\langle 0|)$$
($\rho = |+\rangle\langle+|$)
$$= \frac{1}{3}\cdot 1 + \frac{2}{3}\cdot\frac{1}{\sqrt{2}}\text{Tr}\left(\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 0\\ 1 & 0\end{pmatrix}\right)$$
$$= \frac{1}{3} + \frac{2}{3}\cdot\frac{1}{\sqrt{2}}\cdot\frac{1}{\sqrt{2}} = \frac{2}{3}$$

Mixed states and the Bloch sphere

What about expectation values? For pure states $|\psi\rangle$,

$$\langle X \rangle = \langle \psi | X | \psi \rangle$$

try to reduce to this for pure ρ

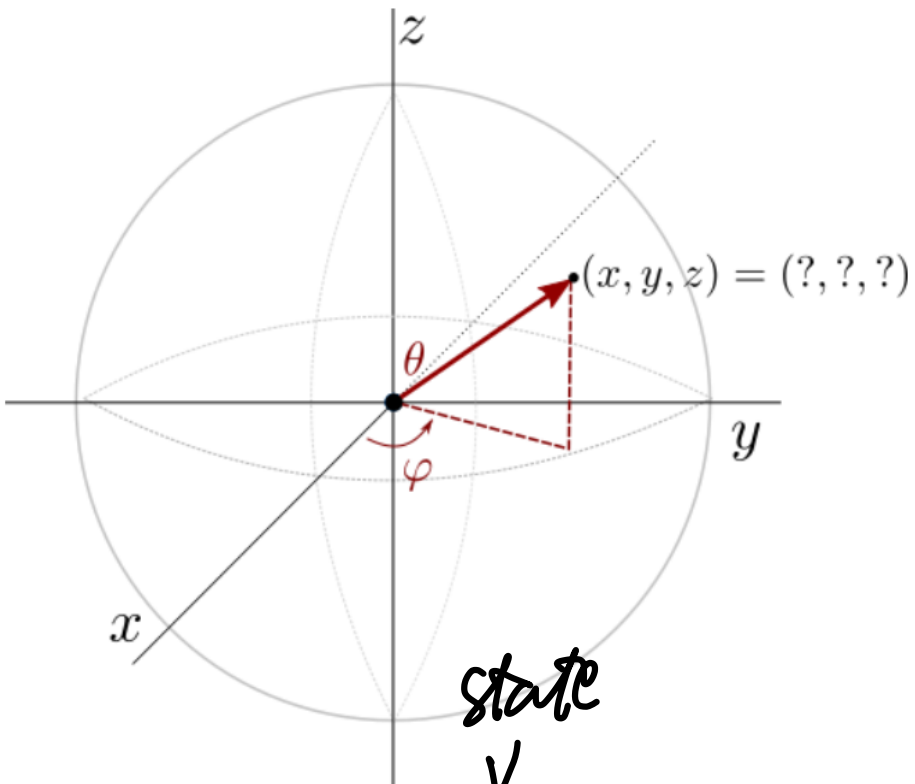
For a mixed state,

$$\begin{aligned}
\langle X \rangle &= 1 \cdot \Pr(+) + (-1) \cdot \Pr(-) \\
&= 1 \cdot \mathrm{Tr}\left(|+\rangle\langle+| \rho\right) + (-1) \cdot \mathrm{Tr}\left(|-\rangle\langle-| \rho\right) \\
&= \mathrm{Tr}\left[|+\rangle\langle+| \rho - |-\rangle\langle-| \rho \right] \\
&= \mathrm{Tr}\left[\left(|+\rangle\langle+| - |-\rangle\langle-| \right) \rho \right] \\
&= \mathrm{Tr}\left[X \rho \right]
\end{aligned}$$

Mixed states and the Bloch sphere

Practice problem set 4...

Write a function that uses quantum measurements to extract its Bloch vector, i.e., its coordinates in 3-dimensional space, as demonstrated by the following graphic:



$\langle X \rangle = \mathrm{Tr}(X\rho)$

$\langle Y \rangle = \mathrm{Tr}(Y\rho)$

$\langle Z \rangle = \mathrm{Tr}(Z\rho)$

state ∨ quantum tomography

Mixed states and the Bloch sphere

Given $\langle X \rangle$, $\langle Y \rangle$, $\langle Z \rangle$, can we determine ρ?

$$\begin{pmatrix} a & b \\ b^* & c \end{pmatrix} \quad a, c \text{ real.}$$

Recall ρ is Hermitian; Paulis are a basis.

$$\rho = \frac{a_I}{2} I + \frac{a_X}{2} X + \frac{a_Y}{2} Y + \frac{a_Z}{2} Z$$

ρ must have trace 1:

$$\mathrm{Tr}(\rho) = \mathrm{Tr}\left[\frac{a_I}{2} I + \frac{a_X}{2} X + \frac{a_Y}{2} Y + \frac{a_Z}{2} Z\right]$$

$$= \mathrm{Tr}\left[\frac{a_I}{2} \cdot I\right] = a_I = 1$$

Trace out another Pauli:

Next time

Content:

- Error channels
- Noise in quantum systems

Action items:

1. Work on project (fill out surveys!)
2. Prepare for quiz 8

Recommended reading:

- From this class: Codebook NT; N & C 2.2.6, 2.4
- For next class: Codebook module NT, D; N & C 8.2-8.3,